

POLYNOMIAL OSCILLATORS IN HEISENBERG PICTURE

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In the quantum-mechanical context, we advocate the use of (integral) Heisenberg equations, the non-numerical solutions of which may be generated by iterations. In an illustrative application to the general (polynomial) oscillators, elementary coefficients result as an additional and unexpected merit of our “generalized Peano-Baker” solutions.

1. Introduction

The Heisenberg picture of quantum systems postulates dynamical equations of the type [1]

$$i \frac{d}{dt} R(t) = [R(t), H], \quad t > 0. \quad (1.1)$$

These relations are currently being used in quantum field theory. In quantum mechanics, on the contrary, the matrix or operator character of observables $R(t)$ (usually, $q(t)$ or $p(t)$) restricts the applicability of (1.1) to a few particular examples [2].

Rather surprisingly, the list of “Heisenberg – solvable” Hamiltonians has recently been complemented by the pair of the phenomenologically appealing “Schroedinger – unsolvable” items [3]

$$H(p, q) = \frac{1}{2}s^{-1}p^{2s} + \frac{1}{4}q^4 \quad (1.2)$$

with $s = 1$ and $s = 2$. Here, in a somewhat different spirit, we shall therefore consider another “realistic” class of Hamiltonians

$$H = \frac{1}{2}p^2 + \frac{1}{2}q^2 + \sum_{m=2}^M \frac{1}{m+1} g_m q^{m+1} \quad (1.3)$$

in Eq. (1.1) and, after an appropriate modification of the standard non-operator differential-equation methods, also succeed in showing its tractability by the non-numerical means.

The presentation of our construction starts from the integral-equation re-formulation of Heisenberg dynamical equations (Sect. 2), followed by the proposal of their

iterative solution (Sect. 3). In Sect. 4, we show that the iterations lead to analytically feasible integrations only. The related formulas and recurrences are displayed in Sect. 5, while Sect. 6 studies the short-time region in more detail. Throughout the text, particular emphasis is laid upon the "one-body" $M = 2$ (quartic) illustrative example, and only the final Sect. 7 also mentions the feasibility of the same construction in systems with more degrees of freedom.

2. Polynomial anharmonicities and the integral Heisenberg equations

Let us first remind the reader that our class of Hamiltonians (1.3) degenerates to the exactly solvable harmonic oscillator for $M = 1$. In this case, an arbitrary initial pair

$$q(0) = Q, \quad p(0) = P, \quad [Q, P] = i \quad (2.1)$$

of the coordinate and impulse operators will evolve in time in accord with Eq. (1.1),

$$\dot{q}(t) = p(t), \quad \dot{p}(t) = -q(t). \quad (2.2)$$

The latter coupled pair of equations possesses the elementary operator solution [3]

$$\begin{aligned} q(t) &= Q \cos t + P \sin t, \\ p(t) &= P \cos t - Q \sin t \end{aligned} \quad (2.3)$$

for all $t > 0$.

In essence, our present methodical proposal is motivated immediately by formula (2.3) or rather by its matrix form

$$\begin{pmatrix} q(t) \\ p(t) \end{pmatrix} = e^{Jt} \begin{pmatrix} Q \\ P \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (2.4)$$

Indeed, for the general Hamiltonian of Eq. (1.3) with $M > 1$ and $H = H\{p(t), q(t)\} = H\{p(0), q(0)\}$, the anharmonic differential analogue of Eq. (2.2) reads

$$\begin{aligned} \dot{q}(t) &= p(t), \\ \dot{p}(t) &= -q(t) - \sum_{m=2}^M g_m [q(t)]^m \end{aligned} \quad (2.5)$$

and may be converted into its integral equivalent

$$\begin{pmatrix} q(t) \\ p(t) \end{pmatrix} = e^{Jt} \begin{pmatrix} Q \\ P \end{pmatrix} - \sum_{m=2}^M g_m \int_0^t dz e^{J(t-z)} \begin{pmatrix} 0 \\ [q(z)]^m \end{pmatrix}. \quad (2.6)$$

in the textbook spirit [4]. Such a nonlinear operator integral Heisenberg equation may then be solved by iterations — this is our basic idea.

3. Iterations

In the context of differential equations, our idea of iteration of their integral counterparts is in fact very old [5]. Up to now, it has found numerous applications in the various (e.g., Riccati-type [6]) problems. Even its linear and non-operator realization (the so called Peano-Baker method [7]) may lead to nontrivial re-arrangements and improvements of convergence of the Taylor or Frobenius power series [8].

In the present paper, the main merit of the proposed iterations will be found in their non-numerical features. Indeed, the lower line in Eq. (2.4) may simply be interpreted as a definition,

$$p(t) = P \cos t - Q \sin t - \sum_{m=2}^M g_m \int_0^t dz \cos(t-z) q^m(z). \quad (3.1)$$

The remaining single (upper) row becomes our fundamental nonlinear integral equation for the operator $q(t)$,

$$q(t) = Q \cos t + P \sin t - \sum_{m=2}^M g_m \int_0^t dz \sin(t-z) q^m(z). \quad (3.2)$$

It may be iterated without any problems.

Immediately, we may notice that the results of iterations will resemble the Ansatz of Refs. [2] and [3] for the first integrals. The time-dependent operator basis has been used there, with the elements

$$\begin{aligned} T_{1,0} &= p(t), \quad T_{0,1} = q(t), \quad T_{2,0} = p^2(t), \\ T_{1,1} &= \frac{1}{2}\{p(t)q(t) + q(t)p(t)\}, \dots \end{aligned} \quad (3.3)$$

The present solutions $q(t)$ and $p(t)$ will be represented in terms of similar elements taken at $t = 0$ (see below).

For the sake of simplicity, let us consider now just the "first nontrivial" $M = 2$ example. It carries all the relevant features of the general case and gives the perturbation-type power-series solutions

$$q(t) = q_0(t) + g q_1(t) + g^2 q_2(t) + \dots \quad (3.4)$$

which may be generated by the simple step-by-step integration,

$$\begin{aligned} q_0(t) &= Q \cos t + P \sin t, \\ 3 q_1(t) &= Q^2(\cos^2 t + \cos t - 2), \\ &+ (PQ + QP) \sin t(\cos t - 1) - P^2(\cos t - 1)^2, \\ 36 q_2(t) &= Q^3[(\cos t - 1)(3 \cos^2 t + 11 \cos t + 16) + 15t \sin t] + \dots, \end{aligned} \quad (3.5)$$

In the purely physical context, the numerical precision of our (truncated) anharmonic-oscillator series (3.4) may be kept under control via our specification of the permissible range of $g = g_2$.

4. The coefficients

The explicit form of components $q_i(t)$ in (3.5) results from a rather complicated integration per partes. In fact, these evaluations may be understood as a sequence of integral transformations

$$\begin{aligned} q_1(t) &= T(g_2 q_0^2 + \dots + g_M q_0^M), \\ q_2(t) &= T\{g_2(q_0 q_1 + q_1 q_0) + \dots + g_M[q_0^{M-1} q_1 + \dots + q_1 q_0^{M-1}]\}, \\ &\dots, \end{aligned} \quad (4.1)$$

each of which is defined by the formula $w(t) = T(v)$ with

$$w(t) = \cos t \int_0^t \sin z v(z) dz - \sin t \int_0^t \cos z v(z) dz. \quad (4.2)$$

In an alternative formulation, we may also write $w(t)$ as a sum of the complex integral

$$R(v) = \frac{e^{-it}}{2i} \int_0^t e^{iz} v(z) dz \quad (4.3)$$

with its hermitean conjugate (h.c.), $w(t) = 2 \operatorname{Re} R(v)$.

The necessary integrations remain elementary. After a few attempts, we find that it is sufficient to know just all the expressions

$$R(z^m e^{inz}) \quad (4.4)$$

with integer n 's and with the non-negative integers m . Indeed, it is easily seen that all the R -transforms (4.4) remain proportional to mere superpositions of expressions appearing in (4.4) as an argument. Thus, all the iterations in (4.1) define the separate components $q_i(t)$ of our solutions $q(t)$ as products of operators Q and P , multiplied by some closed and entirely elementary coefficients. In fact, these coefficients may only be polynomials in sines, cosines and powers of the time-variable t .

The operator part of q_i 's remains similar to the basis used in Ref. [3]. Alternatively, we may also define the $t = 0$ "creation" and "annihilation" operators $A^+ = \frac{1}{2}(Q - iP)$ and $A = \frac{1}{2}(Q + iP)$ and re-write, e.g., the anharmonic-oscillator $M = 2$ result (3.5) in an alternative and more digestible form

$$\begin{aligned} q_0(t) &= A^+ e^{it} + A e^{-it}, \\ 6 q_1(t) &= A^+ A^+ (e^{-it} - 3 e^{it} + 2 e^{2it}) + A A^+ (3 e^{-it} - 6 + 3 e^{it}) + \text{h.c.}, \\ 12 q_2(t) &= A^+ A^+ A^+ e^{-it} (1 - e^{it})^4 + \dots, \\ &\dots \end{aligned} \quad (4.5)$$

A computer with the symbolic manipulation facility will prove extremely useful in this context — all the derivations are rather lengthy while the resulting formulas remain elementary.

5. Recurrences

Due to our knowledge of commutation relations between $p(t)$ and $q(t)$ at both $t = 0$ and $t > 0$,

$$[q(t), p(t)] = i, \quad t \geq 0 \quad (5.1)$$

(and/or between their "creation and annihilation" counterparts), we may easily re-arrange the products of operators in solutions (3.4). Provided that the real basis (products of Q 's and P 's) is preferred [2, 3], a suitable notation

$$V_{rmn}(z) = z^r \sin^m z \cos^n z, \quad W_{rmn}(t) = \int_0^t V_{rmn}(z) dz$$

may be introduced to simplify the formulas.

In the practical computations, the real form of the basis and also the related rules, say, of the type

$$\begin{aligned} W_{rmn} = & \frac{n-1}{m+n} W_{rmn-2} - \frac{r(r-1)}{(m+n)^2} W_{r-2mn} - \frac{rm}{(m+n)^2} W_{r-1m-1n-1} \\ & + \frac{1}{m+n} V_{rm+1n-1} + \frac{r}{(m+n)^2} (V_{r-1mn} - \delta_{r1} \delta_{m0}), \quad n > 0 \end{aligned} \quad (5.2)$$

and/or

$$\begin{aligned} W_{rmn} = & \frac{m-1}{m+n} W_{rm-2n} - \frac{r(r-1)}{(m+n)^2} W_{r-2mn} - \frac{rn}{(m+n)^2} W_{r-1m-1n-1} \\ & + \frac{1}{m+n} V_{rm+1n-1} + \frac{r}{(m+n)^2} V_{r-1mn} \\ & - \frac{1}{m+n} (V_{rm-1n+1} - \delta_{r0} \delta_{m1}), \quad m > 0 \end{aligned} \quad (5.3)$$

with Kronecker deltas δ remain rather clumsy [9]. Nevertheless, the simple initial values

$$W_{r00}(t) = t^{r+1}/(r+1), \quad r \geq 0 \quad (5.4)$$

still enable us to use these formulas as recurrent definitions which specify the integral transformation $v \rightarrow w$ (4.2).

After all, the subtraction form of (4.2) leads to massive cancellations, fortunately. Thus, we may also write

$$T(V_{rpq}) = \sum_{\substack{s, m, n \geq 0 \\ s+m+n \leq M_{rpq}}} d_{smn}^{(rpq)} V_{smn}(t) \quad (5.5)$$

and generate directly the coefficients d from their nonzero initial values

$$\begin{aligned} d_{000}^{(000)} &= -1, \quad d_{001}^{(000)} = 1, \\ d_{010}^{(010)} &= -1/2, \quad d_{101}^{(010)} = 1/2, \quad d_{110}^{(001)} = -1/2. \end{aligned} \quad (5.6)$$

The corresponding (straightforward) re-arrangement of (5.2) and (5.3) need not be written here – let us only note that up to the exceptions listed in (5.6), the value $M_{rpq} = r + p + q$ of the cut-off may be used in (5.5) in all cases with $r + p + q > 0$.

6. Short times

At large values of the couplings $g_2, \dots, g_M$, the iterations need not converge sufficiently quickly while they may still provide useful information about solutions near $t = 0$. Indeed, an inspection of our above $M = 2$ formulas (say, (3.5)) shows that the coefficients exhibit a very specific time-dependence for $|t| \ll 1$. Formally, this enables us to switch [6] to their additional power-series expansions in the variable t .

In a preparatory step, let us introduce some auxiliary and m -dependent sets of “binary” indices $\alpha = \alpha(m)$ and $\beta = \beta(m)$ in the following two arrangements,

$$\begin{aligned}\alpha &= k_m 2^0 + k_{m-1} 2^1 + \dots + k_1 2^{m-1}, \\ \beta &= k_1 2^0 + k_2 2^1 + \dots + k_m 2^{m-1}.\end{aligned}\quad (6.1)$$

Then, we may denote the (mutually not necessarily independent) products of our non-commutative basis operators

$$Q^{k_1}(tP)^{1-k_1} Q^{k_2}(tP)^{1-k_2} \dots Q^{k_m}(tP)^{1-k_m} \quad (6.2)$$

by symbols $X_\alpha(t)$ (or, equivalently, $Y_\beta(t)$) and try to write our solution $q(t)$ in the new mixed-series form

$$q(t) = q_0(t) + \sum_{m=1}^{\infty} (gt^2)^m \sum_{\alpha=0}^{2^{m-1}} \sum_{\beta=\alpha}^{2^{m-1}} Z_{\alpha\beta}(t), \quad (6.3)$$

where

$$\begin{aligned}Z_{\alpha\beta} &= (X_\alpha Y_\beta + X_\beta Y_\alpha) F_{\alpha\beta}^{(0,m)}(t) + gt^2 (X_\alpha Q Y_\beta + X_\beta Q Y_\alpha) F_{\alpha\beta}^{(2,m)}(t) \\ &+ gt^3 (X_\alpha P Y_\beta + X_\beta P Y_\alpha) F_{\alpha\beta}^{(3,m)}(t).\end{aligned}\quad (6.4)$$

An insertion of this Ansatz in our Heisenberg integral equation leads again to an appropriate specification of the new coefficients functions $F_{\alpha\beta}(t)$.

The smooth Taylor-series character of the time-dependence in the new functions $F_{\alpha\beta}(t)$ reads

$$F_{\alpha\beta}^{(n,m)}(t) = \sum_{j=0}^{\infty} c_{\alpha\beta j}^{(n,m)} t^{2j}, \quad n = 0, 2, 3 \quad (6.5)$$

and renders the fully consistent short-time truncations of our anharmonic oscillator solutions possible. In this context, it is also interesting to notice that the “additional” power-series representation (6.5) of our solutions is already ambiguous as a simple Ansatz. Indeed, its insertion in Eqs. (6.4), (6.3) and (3.2) does not suffice for a unique specification of all c ’s. This is not so puzzling – each step of our iteration procedure may simply fix more than one coefficient c at a time.

7. An outlook – two particles

Summatizing our results, the integral Heisenberg equations provide a promising alternative to the standard techniques. In the conclusion, we may predict a new interest in their applicability to the traditional and difficult many-body problems. Preliminarily, the studies of the “next-to-trivial” class of the quasi-harmonic Hamiltonians

$$H_0 = \frac{1}{2} \sum_{i=1}^A (p_i^2 + m_i q_i^2) + \sum_{\substack{i,j=1 \\ i \neq j}}^A G_{ij} q_i q_j \quad (7.1)$$

and/or their various perturbed forms

$$H = H_0 + gV(q_1, q_2, \dots, q_A) \quad (7.2)$$

might provide necessary encouragement and/or inspiration for the future developments.

As a first step in the indicated direction, let us devote a final “ $A > 1$ ” remark to the “first nontrivial” $A = 2$ example with equal masses ($m_1 = m_2 = 1$), variable $G_{12} = c$ (giving the one-body isotropic harmonic oscillator at $c = -1$) and with the related Heisenberg equations

$$\begin{aligned} \dot{q}_1(t) &= p_1(t), \\ \dot{q}_2(t) &= p_2(t), \\ \dot{p}_1(t) &= -q_1(t) - c q_2(t) - g w_1\{q_1(t), q_2(t)\}, \\ \dot{p}_2(t) &= -q_2(t) - c q_1(t) - g w_2\{q_1(t), q_2(t)\} \end{aligned} \quad (7.3)$$

of the type (2.5), where

$$w_i(q_1, q_2) = \frac{d}{dq_i} V(q_1, q_2), \quad i = 1, 2. \quad (7.4)$$

Then, a transition to an integral-equation counterpart to (2.6) may again be based on appropriate generalizations, say,

$$\begin{pmatrix} q_1(t) \\ q_2(t) \\ p_1(t) \\ p_2(t) \end{pmatrix} = e^{Jt} \begin{pmatrix} Q_1 \\ Q_2 \\ P_1 \\ P_2 \end{pmatrix} - g \int_0^t dz e^{J(t-z)} \begin{pmatrix} 0 \\ 0 \\ w_1 \\ w_2 \end{pmatrix} \quad (7.5)$$

with the new auxiliary (and, if needed, partitioned) matrix

$$J = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & -c & 0 & 0 \\ -c & -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & I \\ C & 0 \end{pmatrix}. \quad (7.6)$$

The straightforward algebra enables us to proceed in a very close analogy with the one-particle case — for example, we may arrive at an analogue of solutions (2.3), provided only that we partition

$$\exp(Jt) = \begin{pmatrix} \Lambda C_0 \Lambda & \Lambda C_- \Lambda \\ \Lambda C_+ \Lambda & \Lambda C_0 \Lambda \end{pmatrix} \quad (7.7)$$

and notice that the submatrices.

$$C_0 = \begin{pmatrix} \cos \mu t & 0 \\ 0 & \cos \nu t \end{pmatrix}, \quad C_{\pm} = \begin{pmatrix} \mu^{\pm 1} \sin \mu t & 0 \\ 0 & \nu^{\pm 1} \sin \nu t \end{pmatrix}, \quad (7.8)$$

$$\mu = (1 + c)^{1/2}, \quad \nu = (1 - c)^{1/2}$$

are diagonal due to our special choice of auxiliary factors

$$\Lambda = \Lambda^{-1} = \Lambda^T = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (7.9)$$

The presence of the latter matrices indicates that a transition to the “creation and annihilation” operators might still simplify the resulting formalism significantly.

In a way paralleling our preceding $A = 1$ constructions, the decoupling of equations (into the pair of definitions of p 's and the remaining pair of equations to be iterated for $A = 2$) takes also place for the larger systems. Thus, the same or analogous algebraic manipulations (on a computer) might provide valuable new information on the solutions. Unfortunately, all the related questions (concerning the ways of a “minimal generalization” of exactly solvable harmonic oscillators, etc.) would already lead us far beyond the scope of the present paper.

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